

1 Introduction

In this article I'll write about electromagnetics.

The iter will be:

1. Electric Fields,
2. Capacitors,
3. Dielectrics,
4. Current,
5. Resistors,
6. Magnetic Fields,
7. Electromagnetics.

2 Electric Fields

We'll start by defining what is the force that applies to two charges q_1, q_2 .

Definition 2.1: Force

A **Force** between two charges q_1, q_2 is defined by the relation:

$$F = k_0 \frac{q_1 q_2}{r^2}$$

This definition is in module, so $F = |\vec{F}|$. The force vector is defined by the rule:

$$\vec{F}_{1,2} = k_0 \frac{q_1 q_2}{(\vec{r}_{2,1})^2} \hat{r}_{2,1}$$

This is the force that the charge q_2 applies to the charge q_1 . (the vector $\vec{r}_{2,1}$ is the vector from q_2 to q_1)
It's also to remark that:

$$\vec{F}_{1,2} = -\vec{F}_{2,1}$$

Definition 2.2: Electric Field

The **Field** of a charge q is defined by the limit:

$$\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$$

That definition is equivalent to the more practical:

$$\vec{E}(\vec{r}) = k_0 \frac{q}{r^2} \hat{r}$$

Note. Let's suppose that there are two charges q_1 and q_2 , and we choose a random point in the space p diffent from the two charges. The field in that point will be:

$$\vec{E}_{Tot}(p) = \vec{E}_{q_1}(p) + \vec{E}_{q_2}(p)$$

Note. We use the SI, so the force has [N] as unit, the charge has [C]-oulomb and the field has [N/C] or [V/m].

Proposition 2.1: Sovrapposition

The field in a random point p is given by:

1. For a discrete system of charges:

$$\vec{E}(p) = \sum_{i=1}^N \vec{E}_{q_i}(\vec{r}_i)$$

2. For a continuos system of charges:

$$\vec{E}(p) = \int_Q \vec{E}(\vec{r}) dQ$$

Proof. Just using the note seen before.

□

Example. (Field generated by two charges)

Let's put two charges, q_1 and q_2 , in a plane.

$q_1 = (0, y), q_1 = q, \quad q_2 = (0, -y), q_2 = -q, \quad y > 0, q > 0[C]$.

As you can see, I'll write the position of a charge in the same notation as the value of a charge (when it's clear by the context).

In this example I'll find the value of the field in the point $p = (x, 0), \quad x > 0$.

Let's see the field of the charge q_1 on the point p .

$$\vec{E}_{q_1}(p) = k_0 \frac{q_1}{r_{q_1}^2} \hat{r}_{q_1} = k_0 \frac{q_1}{x^2 + (-y)^2} \left(\frac{x}{\sqrt{x^2 + (-y)^2}}, \frac{-y}{\sqrt{x^2 + (-y)^2}} \right)$$

So we can express the field in the coordinates x, y :

$$\vec{E}_{q_1}(p) = k_0 \left(\frac{q}{(x^2 + y^2)^{3/2}} x, \frac{-q}{(x^2 + y^2)^{3/2}} y \right)$$

The field in p of the charge q_2 is:

$$\vec{E}_{q_2}(p) = k_0 \left(\frac{-q}{(x^2 + y^2)^{3/2}} x, \frac{-q}{(x^2 + y^2)^{3/2}} y \right)$$

So the total field in p is:

$$\vec{E}_{Tot}(p) = \vec{E}_{q_2}(p) + \vec{E}_{q_1}(p) = (0, k_0 \frac{-2q}{(x^2 + y^2)^{3/2}} y)$$

Example. Field generated by a charged string

Let s be an infinite string. I want to describe the field in this system.

Let s be in the z axis. There's an important symmetrical situation.

If the coordinates (x,y) are fixed, the module of the field is independent from z .

furthermore, the module of the field is the same in the circumference of radius r .

That's why it's a good idea to study the field in the coordinates (z,d) where d is the distance from the string.

For every dz there's a charge equals to λdz , where λ is the charge density of the string.

Let's fix a point $p = (0, d)$ in the coordinates (z, d) .

$$d\vec{E}_{dz}(p) = k_0 \frac{\lambda dz}{r_{dz}^3} \vec{r}_{dz}$$

The simmetry of the system tells that the z -contribute of every point z is same in module but different sign of the pont $-z$. For this reason it's resonable to evaluate only the field on the coordinate orizzontaly.

$$d\vec{E}_d(p) = k_0 \frac{\lambda dz}{(z^2 + d^2)^2} \frac{d}{\sqrt{z^2 + d^2}}$$

Now I can evaluate the field integrating on the z axis.

$$\begin{aligned} \vec{E}_d(p) &= \int_{-\infty}^{+\infty} d\vec{E}_d(p) dz \\ &= \int_{-\infty}^{+\infty} k_0 \frac{\lambda d}{(z^2 + d^2)^{3/2}} dz \end{aligned}$$

Note. An elegant algebraic method consists in factoring out z^2 from the denominator:

$$\int \frac{dz}{(z^2 + d^2)^{3/2}} = \int \frac{dz}{z^3 \left(1 + \frac{d^2}{z^2}\right)^{3/2}}$$

By applying the algebraic substitution $u = 1 + \frac{d^2}{z^2}$, we get $du = -\frac{2d^2}{z^3} dz$, which leads to:

$$\begin{aligned} \int \frac{dz}{(z^2 + d^2)^{3/2}} &= -\frac{1}{2d^2} \int u^{-3/2} du \\ &= -\frac{1}{2d^2} \left(\frac{u^{-1/2}}{-1/2} \right) = \frac{1}{d^2 \sqrt{u}} + C \\ &= \frac{z}{d^2 \sqrt{z^2 + d^2}} + C \end{aligned}$$

At this point we have a result

$$\begin{aligned} \vec{E}_d(p) &= [k_0 \lambda d \frac{z}{d^2 \sqrt{z^2 + d^2}}]_{-\infty}^{+\infty} \\ &= k_0 \lambda d \frac{2}{d^2} \\ &= \frac{2k_0 \lambda}{d} \\ &= \frac{\lambda}{2\pi \epsilon_0 d} \end{aligned}$$

Theorem 2.2: Gauss Theorem

The flux of $\vec{E}$ in the empty space of a closed surface S is given by the sum of the charges inside S divided by ϵ_0 .

$$\Phi(\vec{E}) = \frac{Q_{Tot}^{inside}}{\epsilon_0}$$

Using the definitions it becomes:

$$\int_S \vec{E} \hat{n} d\sigma = \int_V \frac{\rho(x)}{\epsilon_0} d\tau$$

Proof. Let's take the elementar part of the flux of a charge inside S .

$$\begin{aligned} d\Phi \vec{E} &= \vec{E} \hat{n} d\sigma \\ &= \vec{E} d\vec{\sigma} \\ &= E \cos\theta d\sigma \\ &= E d\sigma_n \\ &= \frac{Q}{4\pi\epsilon_0} \frac{d\sigma_n}{r^2} \end{aligned}$$

The element of surface is equal to $d\sigma_n = r^2 \sin\theta d\theta d\phi$. So the element of flux is

$$d\Phi \vec{E} = \frac{Q}{4\pi\epsilon_0} \sin\theta d\theta d\phi$$

So the flux of this charge Q will be

$$\begin{aligned} \Phi(\vec{E}) &= \int_S d\Phi(\vec{E}) \\ &= \frac{Q}{4\pi\epsilon_0} \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta \\ &= \frac{Q}{\epsilon_0} \end{aligned}$$

For the principle of Sovraposition the flux is the sum of the flux of the internal charges.

The contribute of the external charges is null due to the fact that the flux that enters the surface is the same of the flux that exits.

□